

From the Big Bang to Life as we know it

This was inspired by Stephen Hawking's 'A brief History of time' which I found extremely fascinating. I decided to write my own summary of how the universe changed over time.

There are four fundamental forces in our universe. The strong, weak, electromagnetic and gravitational forces. As the temperature in a system increases, the strong nuclear force weakens at higher energies gaining asymptotic freedom, while the other forces become stronger. The Electromagnetic and weak forces unify at around 10^{15} Kelvin. At a certain critical temperature - Around 10^{27} Kelvin - the strong, weak and electromagnetic forces reach a point where their strength is equal. This is the point of unification. Due to the unification of these forces at extreme conditions, it could be hypothesised that these forces all encompass a more fundamental force that manifests itself in different ways at lower energy conditions. Gravity is hypothesised to be unified with these forces at around 10^{32} Kelvin.

Consider The Big Bang. We know that our universe is expanding due to the observation of red shift seen across distant galaxies that we observe in all directions. Red shifting is similar to the doppler effect however it occurs due to the expansion of space rather than the motion of galaxies. Suppose you hear a car engine moving away from you, as it moves away, the sound waves emitted from the car must travel an increasingly longer distance while keeping the same velocity. This causes the wavelength of the sound emitted to also increase, meaning the wavelength has been shifted to a larger wavelength. The same phenomenon occurs with Electromagnetic waves - in this case the visible light emitted from distant galaxies. Since this occurs for all distant galaxies, it is implied that all galaxies are moving away from us. It would be very unlikely that we are at the centre of this expansion and so we can assume that it is more probable that all points in spacetime are moving away from one another. If we reverse time, we can conclude that all matter was - at one moment in time - all condensed into one point of no volume and infinite density. It is not understood what happens at this point as there is a breakdown of our current physical model.

Since this point would have infinite density and energy, it would also have an infinite temperature. If everything in the universe was all at one place, it means that all matter would need to have differing momentums or energy states. If not, it would be a violation of the Pauli Exclusion Principle which states that no two fermions can occupy the same space and have the same velocity. Fermions are the fundamental particles of the universe which consist of quarks and electrons. When The Big Bang occurred, the infinite temperature would have created an insanely high pressure, strong enough to overcome the gravitational pull of all mass. After fractions of a second, quarks and gluons could begin to form and the Pauli Exclusion Principle would no longer be violated since these quarks could have different velocities. As the universe expanded, temperature cooled. Approximately 100 seconds after The Big Bang, it is believed that protons and neutrons no longer had sufficient energy to overcome the strong nuclear force, and so protons and neutrons began to fuse to form deuterium nuclei - Heavy hydrogen. It took a much longer time - a few hundred thousand years - for the universe to cool down enough for atoms to be formed by electrons orbiting different nuclei.

Our universe obeys the Heisenberg uncertainty principle which states that as you begin to observe the position of something with increasing accuracy, the accuracy of the measurement of its velocity decreases. This means that you cannot know the exact position and velocity of anything in quantum mechanics. This carries the implication that there should always be some measurable quantum fluctuations throughout every physical field. If not, we would be able to know the velocity of a certain point in space is 0 with total certainty and since we already know the position of this point we chose, the Heisenberg uncertainty principle would be violated.

These quantum fluctuations are extremely small and don't need to be considered with larger scale problems. However, when talking about the big bang, they become rather significant. It is believed that these quantum fluctuations gave rise to variations in the concentration of mass. The places with higher concentration would have a larger gravitational attraction causing them to bundle up together, forming a gas cloud. When the pressure of the universe dropped enough, these hydrogen gas clouds began to collapse due to their gravity. Particles in the gas would become compressed and undergo fusion, releasing energy. This release of energy counteracts the gravitational pull of the gas, and these forces will reach an equilibrium. This formed the first stars in the universe. The hydrogen in the stars acts as a fuel for fusion, preventing the star from collapsing in on itself. Eventually, the star runs out of hydrogen to fuse and starts fusing Helium. Larger stars can fuse elements up to Iron, but after all fusion elements run out the star collapses. When this occurs, all the contents inside the star are spewed out and reimmited back into the universe. This forms more gas clouds which later undergo the same process. However, the heavier elements orbit around the newly formed star since these cannot be fused. This is how we believe that our Solar System was formed. An old star exploding, releasing its contents and forming a new star - Our sun - which gained an orbit of heavier elements.

These elements formed the planets that - at the time - would have been extremely hot. Not an ideal condition for life to form. The Earth is believed to have formed as early as 4.5 billion years ago and it took a few hundred million years for it to cool enough so that oceans began to form. These oceans gave rise to a new environment. Chemical reactions allowed for the production of larger molecules, and some of these molecules had the ability to replicate. In volcanic regions, they would intake Hydrogen Sulfide and release Oxygen, giving rise to a more similar atmospheric content to the present. Sometimes a replicated molecule could contain a mutation making it different. Occasionally, the mutation would affect the replication of a molecule negatively causing it to die out. In other cases, these mutations would come as a benefit to the molecule, allowing it to replicate quicker and survive more efficiently. This would lead to the beginning of evolution and sometime after this, the evolutionary offspring could have the necessary complexity to be deemed 'living'. Allowing this slow evolutionary process to iterate over millions of years gives rise to more complex life forms existing. Eventually we evolve to become humans and gain consciousness. In the grand scheme of things, we are greatly insignificant in the history of the universe and its grandeur.